

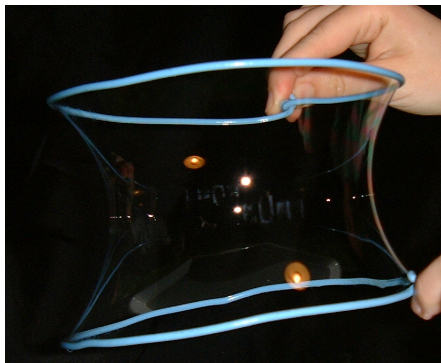
Morse Indices and Physics-Informed Neural Networks

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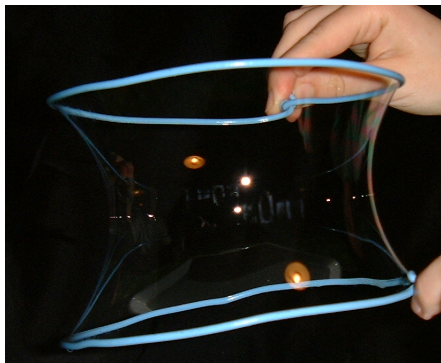
Minimal Surfaces



1

¹link to the original image

Minimal Surfaces



1

Mean curvature is zero.

¹link to the original image

Minimal Surfaces



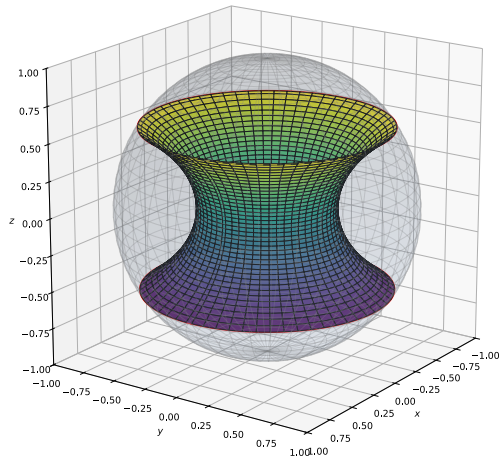
2

Parametric equation

$$(c \cosh(t) \cos u, c \cosh(t) \sin u, ct)$$

for $u \in [0, 2\pi)$, $t \in \mathbb{R}$, and c is non-zero real constant.

Free Boundary Minimal Surfaces



Mathematical Setup

Morse index and nullity

The *Morse index* of Σ is the maximal dimension of a subspace $W \subseteq C^\infty(\Sigma)$ on which the index form S is negative definite; that is,

$$\text{Ind}(\Sigma) = \max \{ \dim W : W \subseteq C^\infty(\Sigma), S(u, u) < 0 \text{ for all } u \in W \setminus \{0\} \}.$$

The *nullity* of Σ is the dimension of the null space of S , i.e. the space of $u \in C^\infty(\Sigma)$ for which $S(u, v) = 0$ for every $v \in C^\infty(\Sigma)$.

Mathematical Setup

Theorem [4, 1]

Let $\Sigma^k \subset \Omega^{k+1}$ be a smooth, properly immersed, orientable FBMS for which $h^{\partial\Omega}(\nu, \nu) = -c$ is constant. Then, counting multiplicities,

1. the Morse index of Σ equals the number of negative eigenvalues of the fixed-boundary problem plus the number of eigenvalues of the Dirichlet-to-Neumann map that are strictly less than c ; and
2. the nullity of Σ equals the dimension of the eigenspace associated with the eigenvalue c .

What is we are going to solve?

$$\begin{cases} f''(t) + \left(\frac{2}{\cosh^2 t} - n^2 \right) f(t) = 0, & t \in (-T, T), \\ Tf'(+T) = \delta f(+T), \quad -Tf'(-T) = \delta f(-T). \end{cases}$$

Physics-Informed Neural Network Method - Architecture

[2, 3]

Component	Specification	Notes
Input	$t \in [-T, T]$	one collocation coordinate
Hidden layers	4×64 , tanh	fully connected
Parity	$f = N(t) \pm N(-t)$	exact even/odd by construction
Eigenvalue	δ (trainable scalar)	learned by gradient descent
Optimizer	Adam, $\text{lr} = 2 \times 10^{-3}$	cosine annealing to 10^{-5}
Collocation	600 uniform points	interior of $[-T, T]$
Epochs	9000–15000	per mode

Table: PINN architecture and training configuration for the one-dimensional Robin eigenvalue problem.

Loss Function

The total loss is $\mathcal{L} = \mathcal{L}_{\text{pde}} + 5\mathcal{L}_{\text{bc}} + 2\mathcal{L}_{\text{norm}}$, with

$$\mathcal{L}_{\text{pde}} = \frac{1}{M} \sum_{i=1}^M \left[f''(t_i) + \left(\frac{2}{\cosh^2 t_i} - n^2 \right) f(t_i) \right]^2,$$

$$\mathcal{L}_{\text{bc}} = [Tf'(+T) - \delta f(+T)]^2 + [-Tf'(-T) - \delta f(-T)]^2,$$

$$\mathcal{L}_{\text{norm}} = \left(\int_{-T}^T f^2 dt - 1 \right)^2,$$

where the $\{t_i\}_{i=1}^M$ are the $M = 600$ collocation points.

Recovered Eigenvalues

Mode	Exact δ	PINN δ	error	PDE resid	$\delta < 1?$
$n = 0$ odd	+0.43923	+0.43923	2.6×10^{-6}	6.9×10^{-5}	Yes ($\times 1$)
$n = 1$ even	-1.00000	-0.99990	9.7×10^{-5}	5.2×10^{-4}	Yes ($\times 2$)
$n = 1$ odd	+1.00000	+1.00000	1.2×10^{-6}	2.8×10^{-4}	No (null.)
$n = 2$ even	+2.00000	—	—	—	No
$n = 3$ even	+3.42087	—	—	—	No

Table: Jacobi-Steklov eigenvalues of the critical catenoid.

PINN Verification

PINN verification: Morse index of the critical catenoid

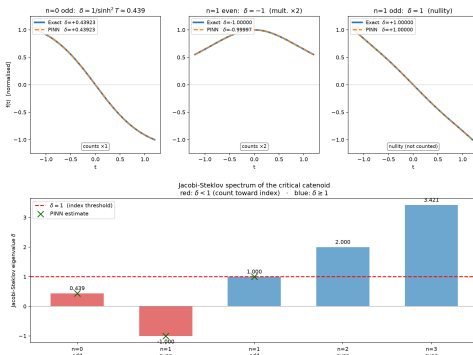


Figure: Top: analytic (solid) versus PINN (dashed) eigenfunctions for the three lowest Jacobi-Steklov modes. The $n = 0$ odd mode ($\delta \approx 0.439$) and the $n = 1$ even mode ($\delta = -1$, multiplicity 2) lie below the threshold $\delta = 1$ and count toward the index; the $n = 1$ odd mode ($\delta = 1$) is the nullity mode. Bottom: the Jacobi-Steklov spectrum for $n = 0$ to 3 ; red bars mark $\delta < 1$, blue bars $\delta \geq 1$, and crosses the PINN estimates.

Future plans...

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